Clinical and Hormonal Predictors of BMI in PCOS/PMOS: A Statistical Analysis and Dataset Coverage Audit

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Abstract—The 2026 renaming of polycystic ovary syndrome (PCOS) to polyendocrine metabolic ovarian syndrome (PMOS) emphasizes that the condition is endocrine, metabolic, reproductive, dermatological, and psychological. We analyze a widely used Kaggle PCOS dataset to ask which measured variables are associated with BMI among PCOS-positive participants and whether the dataset contains the biosignals needed for PMOS-era metabolic modeling. The analysis uses descriptive statistics, simple regression, Welch's two-sample t-test, one-way ANOVA, multiple linear regression, adjusted R^2, VIF, residual diagnostics, and a construct coverage audit. Self-reported weight gain is the dominant BMI-associated predictor; the available laboratory block explains almost none of the BMI variation. We interpret this as a dataset limitation, not evidence against metabolic mechanisms in PMOS.

I. Introduction

PCOS was renamed PMOS in May 2026 to reduce the misleading emphasis on ovarian cysts and to foreground endocrine and metabolic features [1], [2]. This creates a useful statistical question: does a popular PCOS dataset measure the domains now emphasized by PMOS? Our original course question remains BMI-focused: among PCOS-positive participants, which clinical, hormonal, lifestyle, and ultrasound variables are associated with BMI?

II. Data and Methods

We used the Kaggle PCOS dataset by Kottarathil [3]. The main workbook has 541 rows and 45 columns; the BMI analysis is restricted to 177 PCOS-positive rows. A second infertility file was audited separately and appears to contain the same patients with patient IDs offset by 10000 and duplicate AMH/beta-HCG fields, so it does not add fasting insulin, testosterone, fasting glucose, or HOMA-IR.

BMI was treated as a numerical response. We inspected its distribution, fit a simple linear regression with waist-to-hip ratio, used a Welch t-test for cycle regularity, used one-way ANOVA across total follicle-count groups, and fit multiple regression models with clinical, laboratory, ultrasound, and combined predictor blocks. Multicollinearity was assessed by VIF and model adequacy by residual and Q-Q plots.

III. Results

The BMI distribution was only mildly right-skewed. Waist-to-hip ratio had a weak, non-significant simple linear association with BMI. The Welch test suggested higher mean BMI among participants with irregular cycles, but this difference was reduced after adjustment. Follicle-count group did not show a significant ANOVA result, so Tukey post-hoc comparisons were not warranted.

TABLE I
Primary STAT 501 results.

Analysis	Result
BMI distribution	mean 25.47, SD 4.40, skew 0.232
Simple regression	WHR slope 7.93, R2 0.007, p 0.2678
Welch t-test	irregular mean 26.28 vs regular 24.53, p 0.0076
One-way ANOVA	follicle groups F 0.10, p 0.9039
Multiple regression	full adj. R2 0.237; clinical adj. R2 0.260; weight gain p <0.001

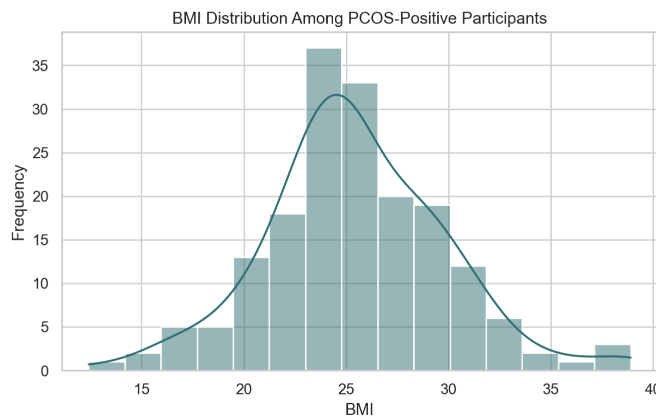

Fig. 1. BMI distribution among PCOS-positive participants.

The clinical-only block had the strongest adjusted R^2 among compact models. The laboratory-only model had adjusted $R^2 = -0.008$, indicating that available labs such as AMH, LH/FSH, TSH, PRL, and RBS did not explain BMI better than an intercept-only baseline. In the full model, non-intercept VIF values were close to 1, so multicollinearity was not the explanation. Residual diagnostics showed no major pattern that would overturn the main exploratory conclusion.

IV. PMOS Coverage Audit

The dataset is strongest on reproductive and ovarian structure variables, partial on endocrine and dermatological symptoms, weak on metabolic/cardiovascular variables,

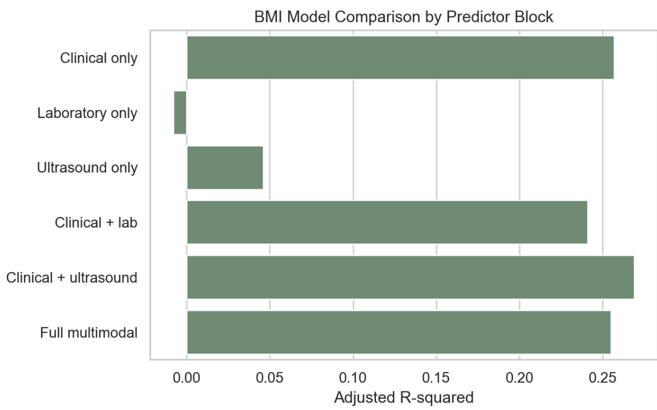

Fig. 2. Adjusted R^2 by BMI predictor block.

TABLE II
BMI model comparison by predictor block.

Model block	n	Adj. R2	AIC
Clinical only	176	0.257	976.200
Laboratory only	176	-0.008	1029.000
Ultrasound only	176	0.046	1019.300
Clinical + lab	176	0.241	984.700
Clinical + ultrasound	176	0.269	978.000
Full multimodal	176	0.255	986.000

and absent on psychological or quality-of-life measures. Therefore, the weak laboratory BMI result should not be read as evidence that endocrine or metabolic mechanisms are unimportant. Instead, the dataset lacks the main metabolic biosignals emphasized by PMOS: fasting insulin, fasting glucose, HOMA-IR, OGTT, lipids, and detailed androgen measures.

V. Discussion

The most consistent BMI-associated variable was self-reported weight gain. This is plausible as a broad clinical signal, but it is not a mechanistic biomarker. The analysis is cross-sectional and observational, so it estimates association rather than causation. The main value of this project is a statistical and measurement critique: a widely used PCOS dataset can support exploratory BMI modeling and PCOS-status classification, but it is structurally limited for PMOS-era metabolic mechanism analysis.

VI. Conclusion

For the course question, clinical variables outperformed the available laboratory variables for BMI prediction among PCOS-positive participants. For the PMOS-era interpretation, the result highlights a data gap: future datasets should include fasting insulin, fasting glucose, HOMA-IR, lipids, blood pressure, androgen assays, and validated mental-health measures to represent PMOS as a multisystem condition.

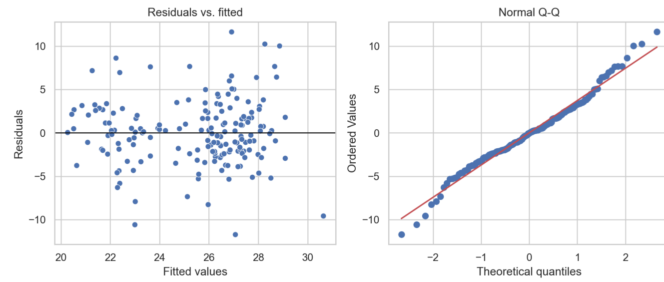

Fig. 3. Residual and Q-Q diagnostics for the multiple regression.

TABLE III
PMOS construct coverage in the Kaggle variables.

PMOS domain	Coverage	Major gaps
Reproductive / ovarian	Strong	diagnostic visit notes and formal phenotype labels
Endocrine	Partial	testosterone, SHBG, DHEAS, free androgen index
Metabolic / cardiovascular	Weak proxy	fasting insulin, fasting glucose, HOMA-IR, OGTT, lipids
Dermatological / symptoms	Partial	standardized hirsutism/acne scales
Psychological / quality of life	Absent	depression, anxiety, sleep, stigma, quality-of-life scales

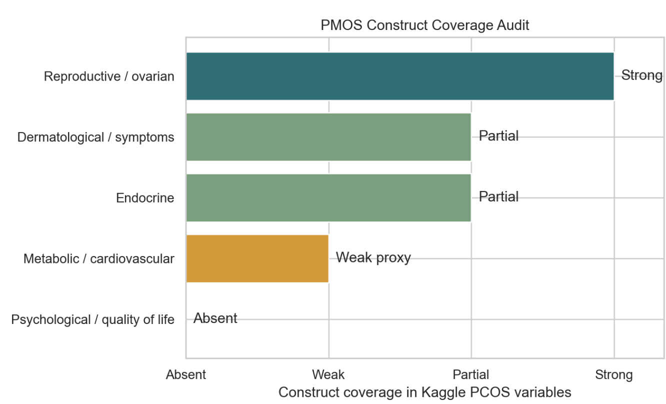

Fig. 4. Construct coverage audit under the PMOS framing.

References

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